

# KAMYAB AZIZI

## CONTACT

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- ☐ HomePage: <https://kamyabazizi.github.io>
- ☐ GitHub: <https://github.com/kamyabazizi>
- ☐ Linked-in: <https://linkedin.com/in/kamyab-azizi-684236167>
- ☐ Google Scholar: <https://scholar.google.com/citations?hl=en&user=BUCAMJ8AAAAJ>

## CURRICULUM VITAE

### INTERESTS:

Machine Learning  
Deep Learning  
Signal Processing  
Embedded Systems  
Computer Vision  
HW/SW co-design

### EDUCATION:

**Master study at Amirkabir University of Technology-Electrical and Electronic engineering (Tehran Polytechnic)**  
**Thesis Title:** Transfer learning in pruned deep neural networks based on the lottery ticket hypothesis  
**Supervisor:** Dr. Hassan Taheri  
2020-2022 GPA 17.25/20 (3.77/4 GPA)

**BSc at Amirkabir University of Technology-Electrical and Electronic Engineering (Tehran Polytechnic)**  
**Thesis Title:** Image compression using hybrid methods based on digital image processing and linear algebra  
**Supervisor:** Dr. Hassan Taheri  
2015 -2020 GPA 16.07/20 (3.2/4 GPA)

### HOBBIES:

Football  
Movies  
Reading Books  
Swimming

### EXPERIENCE:

**Autonomous Driving Algorithm Engineer at Software Motion Co.**  
Working on ADAS function based on AUTOSAR  
Since Apr. 2023

**Research Assistant at AUT**  
Amirkabir University of Technology-Electrical and Electronic Engineering (Tehran Polytechnic) – (Oct. 2020-Dec. 2022)

**Volunteer Student Committee at the 5<sup>th</sup> International Conference on Robotics and Mechatronics (ICROM)**  
Robotic Society of Iran (RSI) – Oct 2017

**Electronics Engineer – Internship at TOOBAL Engeering Co.**  
Working on Programmable Logic Device (PLD) – Summer 2017

### SKILLS:

#### Programming Skills:

- Python Programming Language
- MATLAB & SIMULINK
- VHDL Programming-FPGA
- C/C++
- SQL

#### Knowledge and Theoretical Skills:

- AI-Machine Learning-Deep Learning
- Image-Signal-Speech processing
- Pattern Recognition-Computer Vision
- Logic Circuits and Computer Architecture

#### Tools and Technologies:

- AVR (Codevision) & ARM (Keil)
- Microprocessors and assembly language
- Linux
- LaTeX
- ROS

#### Frameworks:

- Tensorflow and Keras
- Pytorch
- Pyspark
- NumPy and Pandas

### LANGUAGES:

|                                                      |                                           |                                                     |                                                     |
|------------------------------------------------------|-------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|
| » <b>English</b><br>Professional working proficiency | » <b>German</b><br>Elementary proficiency | » <b>Kurdish</b><br>Native or bilingual proficiency | » <b>Persian</b><br>Native or bilingual proficiency |
|------------------------------------------------------|-------------------------------------------|-----------------------------------------------------|-----------------------------------------------------|

### SELECTED PROJECTS:

#### Projects (EE Dept. of AUT & SW co.):

|      |                                                                                                                                                     |                    |
|------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| 2025 | Breast Cancer Classification<br><i>Classify benign and malignant cancer image in BreakHis dataset using ResNeXt50 model</i>                         | Pytorch            |
| 2024 | L2 and L4 Auto-Driving Vehicle<br><i>Integrates multiple advanced technologies and real-time environmental data acquisition for safe navigation</i> | C/C++, ROS, MATLAB |
| 2022 | Deep Learning [Dr. Faez]<br><i>Audio-based drone or Unmanned Aerial Vehicle detection and identification using Deep Learning</i>                    | Keras              |
| 2021 | Computer Vision [Dr. Faez]<br><i>Map to Aerial Image Translation</i>                                                                                | Pytorch            |
| 2021 | Neural Networks [Dr. Faez]<br><i>Music genres classification task with Convolutional Neural Networks</i>                                            | Keras              |

|      |                                                                                                                                                                     |                                |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|
| 2021 | Big Data Analytics [Dr. Sharifian]<br><i>Malware classification in BIG 2015 dataset for Microsoft challenge with the decision tree and Random forest classifier</i> | Pyspark                        |
| 2020 | Statistical Pattern Recognition [Dr. Faez]<br><i>Singular value decomposition algorithm for octonion signal and image denoising</i>                                 | MATLAB                         |
| 2020 | Data Analytics [Dr. Sharifian]<br><i>Data acquisition and processing environment for IoT applications with Apache Kafka and TensorFlow Serving</i>                  | Virtual Machine, Linux, Docker |
| 2019 | FPGA [Dr. Sharifian]<br><i>Implementing pipe-lined CORDIC Block on FPGA</i>                                                                                         | VHDL                           |

### CERTIFICATES:

|      |                                                                                                                                                 |                                                                                                                                                                                                 |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 | <b>Convolutional Neural Networks</b><br><i>DeepLearning.AI</i>                                                                                  | <a href="https://www.coursera.org/account/accomplishments/verify/VKBSAVYUZSSS">https://www.coursera.org/account/accomplishments/verify/VKBSAVYUZSSS</a>                                         |
| 2022 | <b>Generative Adversarial Networks (GANs) Specialization</b><br><i>DeepLearning.AI</i>                                                          | <a href="https://www.coursera.org/account/accomplishments/specialization/K68AT2RDD8XN">https://www.coursera.org/account/accomplishments/specialization/K68AT2RDD8XN</a>                         |
| 2022 | <b>Structuring Machine Learning Projects</b><br><i>DeepLearning.AI</i>                                                                          | <a href="https://www.coursera.org/account/accomplishments/verify/ZUP5P4NBM3NX">https://www.coursera.org/account/accomplishments/verify/ZUP5P4NBM3NX</a>                                         |
| 2022 | <b>Machine Learning</b><br><i>Stanford University</i>                                                                                           | <a href="https://www.coursera.org/account/accomplishments/verify/NRHF4N5YMFH2">https://www.coursera.org/account/accomplishments/verify/NRHF4N5YMFH2</a>                                         |
| 2022 | <b>Improving Deep Neural Networks: Hyperparameter Tuning, Regularization, and Optimization</b><br><i>DeepLearning.AI</i>                        | <a href="https://www.coursera.org/account/accomplishments/verify/ZVHDKSYV5FVZ">https://www.coursera.org/account/accomplishments/verify/ZVHDKSYV5FVZ</a>                                         |
| 2022 | <b>Neural Networks and Deep Learning</b><br><i>DeepLearning.AI</i>                                                                              | <a href="https://www.coursera.org/account/accomplishments/verify/UF77DE6SVWZD">https://www.coursera.org/account/accomplishments/verify/UF77DE6SVWZD</a>                                         |
| 2021 | <b>FIFTH IPM ADVANCED SCHOOL ON COMPUTING &amp; ARTIFICIAL INTELLIGENCE</b><br><i>IPM Advanced School on Computing: Artificial Intelligence</i> | <a href="https://raw.githubusercontent.com/kamyabazizi/kamyabazizi.github.io/main/images/CER2.jpg">https://raw.githubusercontent.com/kamyabazizi/kamyabazizi.github.io/main/images/CER2.jpg</a> |
| 2017 | <b>Executive Member in the 5<sup>th</sup> RSI International Conference on Robotics and Mechatronics</b><br><i>ICRoM</i>                         | <a href="https://raw.githubusercontent.com/kamyabazizi/kamyabazizi.github.io/main/images/CER1.jpg">https://raw.githubusercontent.com/kamyabazizi/kamyabazizi.github.io/main/images/CER1.jpg</a> |

### HONORS:

- 15<sup>th</sup> rank in the Iranian National Scientific Olympiad for University Students in Electrical and Electronics Engineering  
*Issued by Sanjesh Organization – Dec 2020* (<https://gto.aut.ac.ir/content/8449/>)
- Ranked Within the Top 0.27%  
*(30<sup>th</sup>) Amongst ~11,000 participants in the national M.Sc. Entrance Exam in Electrical Engineering – Sep 2020*
- Ranked Within the Top 0.25% of students in the Iranian National University Entrance Exam  
*(454<sup>th</sup>) Amongst ~181,000 participants in the national university entrance exam – Sep 2015*

### NAME AND EMAIL OF TWO REFERENCES:

**HASSAN TAHERI.** Associate Professor, Department of Electrical Engineering, Amirkabir University of Technology

Email: [htaheri@aut.ac.ir](mailto:htaheri@aut.ac.ir)

**KARIM FAEZ.** Professor, Department of Electrical Engineering, Amirkabir University of Technology

Email: [kfaez@aut.ac.ir](mailto:kfaez@aut.ac.ir)

## KEY PUBLICATIONS

**Using Structured Pruning to Find Winning Lottery Tickets:** (<https://ieeexplore.ieee.org/abstract/document/10105376>)

**Fabrication of heartbeat signal acquisition device and monitoring of the signal on mobile screen:** (<https://arxiv.org/abs/2302.06272>)

**Convolutional Neural Network Classifier for Unmanned Aerial Vehicles Detection and Identification Using Mel-Frequency Spectrograms** (<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/tje2.70135>)

**Small Dataset Machine Learning Approach in Nanoparticle Synthesis Experiments** (Under review by International Journal of Precision Engineering and Manufacturing-Green Technology)

**Robust Breast Cancer Classification Across Magnifications Using Fine-Tuned ResNeXt50\_32x4d with Hybrid Loss** (Under review by Biomedical Signal Processing & Control >>>> Preprint: [https://papers.ssrn.com/sol3/papers.cfm?abstract\\_id=5529538](https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5529538))

**Research Trends in Cloud Computing-Based Signal Processing: A Bibliometric Analysis**  
(<https://ieeexplore.ieee.org/document/11551752>)

**AI-Enhanced Signal Processing in Biomedical Devices: A Comprehensive Review** (Under review by Elsevier’s Signal Processing journal)